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DEHUMIDIFIER

Abstract

A dehumidifier according to an embodiment may include a housing including a suction grill formed on a rear surface thereof, and a discharge port formed on an upper surface thereof; a dehumidification module provided inside of the housing; a water tank detachably provided on one side of the housing and that stores condensate generated during a dehumidification process; a discharge cover that opens and closes the discharge port; a tilting module connected to a bottom of the discharge cover inside of the housing to tilt the discharge cover; and a shielding cover that accommodates the tilting module. The dehumidification module may include at least a compressor, a condenser, an evaporator, and a fan module. The tilting module may include a tilting motor fixed to the shielding cover; a tilting gear connected to a shaft of the tilting motor; and a tilting neck curved at a predetermined curvature and having on one surface thereof a tilting rack that engages with the tilting gear.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the benefits of priority to Korean Patent Application No. 10-2024-0026073, filed in Korea on Feb. 22, 2024, the contents of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

[0002] A dehumidifier is disclosed herein.

2. Background

[0003] A dehumidifier is a device that lowers a humidity in a space in which the dehumidifier is installed by suctioning in air, removing moisture contained in the air, and discharging the dehumidified air back into the room. In general, dehumidifiers are widely used to lower the humidity of indoor air in humid summer or to quickly dry laundry.

[0004] Dehumidifiers are divided into a desiccant type and a condensation type depending on a dehumidification method. In the condensation type, a refrigerant cycle is installed inside of the dehumidifier, air suctioned in by a fan passes through an evaporator to condense moisture contained in the air, and then passes through the condenser and is discharged back into a room in a heated state.

[0005] Korean Patent Publication No. 2015-0124834, which is hereby incorporated by reference, discloses a dehumidifier having a disadvantage in that a discharge port is always exposed to the outside, which is not only unattractive in appearance, but also causes foreign substances, such as dust, to flow into a fan module through the discharge port when not in use. In addition, as a flow direction of air discharged into a room through the discharge port is fixed in a vertical direction, there is a disadvantage that the dehumidifier itself must be rotated to adjust the flow direction in lateral directions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:

[0007] FIG. 1 is a perspective view of a dehumidifier according to an embodiment with a discharge port in a closed state;

[0008] FIG. 2 is a perspective view of the dehumidifier of FIG. 1 with the discharge port in an open state;

[0009] FIG. 3 is a rear perspective view of the dehumidifier of FIG. 1;

[0010] FIG. 4 is a perspective view of the dehumidifier of FIG. 1 with a housing removed;

[0011] FIG. 5 is a front perspective view of a cover drive unit according to an embodiment;

[0012] FIG. 6 is a rear perspective view of the cover drive unit of FIG. 5;

[0013] FIG. 7 is a perspective view showing the cover drive unit with a cover sleeve removed;

[0014] FIG. 8 is a longitudinal cross-sectional view of the cover drive unit, taken along line VIII-VIII of FIG. 7;

[0015] FIGS. **9** and **10** are perspective views of a tilting module according to an embodiment;
[0016] FIG. **11** is a perspective view showing a combination of a shielding cover and a tilting module according to an embodiment;
[0017] FIG. **12** is a longitudinal cross-sectional view, taken along line XII-XII of FIG. **11**;
[0018] FIG. **13** is a longitudinal cross-sectional view, taken along line XIII-XIII in FIG. **11**; and
[0019] FIG. **14** is an enlarged view of portion A of FIG. **8**.

DETAILED DESCRIPTION

[0020] Hereinafter, a dehumidifier according to an embodiment will be described with reference to the drawings.

[0021] FIG. **1** is a perspective view of a dehumidifier according to an embodiment with a discharge port in a closed state. FIG. **2** is a perspective view of the dehumidifier of FIG. **1** with the discharge port in an open state. FIG. **3** is a rear perspective view of the dehumidifier of FIG. **1**. FIG. **4** is a perspective view of the dehumidifier of FIG. **1** with a housing removed.

[0022] Referring to FIGS. **1** to **4**, dehumidifier **10** according to an embodiment may have a slim shape with a relatively thin thickness in a frontward-to-backward direction. The dehumidifier **10** may have a thickness extending in the frontward-to-backward direction, a width extending in a leftward-rightward or lateral direction, and a height extending in a vertical direction.

[0023] The dehumidifier **10** may include a housing that forms an exterior thereof. The dehumidifier **10** may further include a dehumidification module installed inside of the housing. The dehumidifier **10** may further include a water tank **20** that stores condensate generated during the dehumidification process.

[0024] The housing may include a base **11**, a case **12**, and a top cover **13**. The base **11** may form a lower portion of the dehumidifier **10**. A plurality of wheels **16** may be mounted on a bottom of the base **11** to facilitate movement of the dehumidifier **10**. The case **12** may be disposed on an upper edge of the base **11**, and an upper opening of the case **12** may be shielded by the top cover **13**.

[0025] A suction grill **121** may be formed on a back of the case **12** to intake indoor air. A filter module may be mounted in front of the suction grill **121**. The suction grill **121** may be defined as a suction port or inlet through which indoor air is suctioned.

[0026] A main discharge port **131**, through which dehumidified air may be discharged may be formed on one or a first side of the top cover **13**. The main discharge port **131** may be selectively opened or closed by the discharge cover **14**.

[0027] A handle **15** may be provided on the other or a second side of the top cover **13**. The handle **15** has an n-shape so that a vertical portion thereof may be inserted into the case **12** or pulled out from the case **12**. When the vertical portion is fully inserted into the case **12**, a horizontal portion of the handle **15**, that is, an upper surface of a portion configured to be held by a hand, may be flush with the top cover **13**.

[0028] The main discharge port **131** may be disposed at one or a first edge of the top cover **13**. The handle **15** may be placed on the other or a second edge of the top cover **13**. The main discharge port **131** and the handle **15** may be located on opposite edges. A discharge grill **141** may be mounted on the main discharge port **131**.

[0029] A cutout into which the water tank **20** may be inserted may be formed at one edge of the case **12**, and the water tank **20** may be detachable from the case **12**. When the water tank **20** is mounted on the case **12**, an outer surface of the water tank **20** may form a portion of the case **12**. The water tank **20** may be located right beneath the main discharge port **131**.

[0030] The dehumidification module may include a compressor **31** that compresses a refrigerant, a condenser **32** that condenses the refrigerant that has passed through the compressor **31**, an expansion valve that expands the refrigerant that has passed through the condenser **32** into a two-phase refrigerant, and an evaporator **33** that evaporates the refrigerant that has passed through the expansion valve. The dehumidification module may further include a fan module **34** that suctions indoor air through the suction grill **121**. The evaporator **33** may be located in front of the suction

grill **121**, the condenser **32** may be located in front of the evaporator **33**, and the fan module **34** may be located in front of the condenser **32**.

[0031] The condenser **32** and the evaporator **33** may be seated on an upper surface of a drain pan **35**. Condensed water flowing down from a surface of the evaporator **33** may flow along the drain pan **35** and collected in the water tank **20**.

[0032] An outlet of the fan module **34** may be connected to the main discharge port **131**. The discharge grill **141** may be disposed right above the outlet of the fan module **34**.

[0033] Indoor air may sequentially pass through the suction grill **121**, a filter module, the evaporator **33**, and the condenser **32** and then be suctioned into the fan module **34**. Air suctioned into the fan module **34** may be discharged through the main discharge port **131**.

[0034] The air introduced into the case **12** through the suction grill **121** may be dehumidified while passing through the evaporator **33**, and moisture contained in the air condensed. As a result, the incoming air becomes low-temperature and dry as it passes through the evaporator **33**.

[0035] The low-temperature dry air absorbs heat while passing through the condenser **32**, and as a result, is changed to a high-temperature dry state and is then discharged back into the room through the main discharge port **131**. The term “high temperature air” may refer to air with a temperature slightly higher than a temperature of an indoor space in which the dehumidifier **10** is installed.

[0036] A control box **17** may be accommodated inside of the case **12**, to control operation of the dehumidification module and various drive means.

[0037] A sub discharge port **122** through which dehumidified air may be discharged may be further formed in the case **12**. The sub discharge port **122** may be selectively opened or closed by a damper module **40**.

[0038] When the sub discharge port **122** is open, the main discharge port **131** may be closed, and when the sub discharge port **122** is closed, the main discharge port **131** may be open. Accordingly, the dehumidified air may be selectively discharged only through either the main discharge port **131** or the sub discharge port **122**.

[0039] The sub discharge port **122** may be formed on a front surface or a rear surface of the case **12**. The sub discharge port **122** may be disposed at an upper side of the fan module **34**. For example, the sub discharge port **122** may be formed on the rear surface of the case **12** and may be located above the suction grill **121**.

[0040] The dehumidifier **10** may further include the damper module **40** that selectively opens or closes the sub discharge port **122**. The damper module **40** may be installed inside of the sub discharge port **122** and may open or close the sub discharge port **122**. The damper module **40** may operate in a direction to close the sub discharge port **122**.

[0041] A cover drive unit **50** according to an embodiment may be mounted on or at the outlet of the fan module **34**. The cover drive unit **50** may perform an operation of tilting the discharge cover **14** in upward and downward directions and an operation of rotating the discharge cover **14** in a tilted state clockwise or counterclockwise.

[0042] FIG. **5** is a front perspective view of a cover drive unit according to an embodiment. FIG. **6** is a rear perspective view of the cover drive unit of FIG. **5**.

[0043] Referring to FIGS. **5** and **6**, the cover drive unit **50** according to an embodiment may include a guide plate **51** mounted on or at the outlet of the fan module **34**; a cylindrical cover sleeve **54** with a rotation rack **541** formed at a lower end of an outer peripheral surface thereof and mounted on the guide plate **51**; a pinion **53** mounted on the guide plate **51** to be engaged with the rack gear **541**; a rotation motor **52** that supplies a rotational force to the pinion **53**; and a tilting module that tilts the discharge cover **14** in the upward and downward directions.

[0044] The discharge cover **14** may selectively block an opening formed at a top of the cover sleeve **54**, and an inner diameter of the cover sleeve **54** may be formed to be the same in size as an inner diameter of the main discharge port **131**. In addition, the discharge grill **141** may be coupled to an inside of an upper end of the cover sleeve **54**, may rotate as one body with the cover sleeve

54, and may be opened or closed by the discharge cover **14**. A plurality of fastening holes **542** may be formed on a rear surface of the cover sleeve **54**.

[0045] FIG. **7** is a perspective view showing the cover drive unit with a cover sleeve removed. FIG. **8** is a longitudinal cross-sectional view of the cover drive unit, taken along VIII-VIII in FIG. **7**. FIGS. **9** and **10** are perspective views of a tilting module.

[0046] Referring to FIGS. **7** to **10**, the cover drive unit **50** according to an embodiment may include the guide plate **51**; a shielding cover **55** coupled to a rear surface of the cover sleeve **54**; and a tilting module accommodated inside the shielding cover **55**. An outlet **512** having a size corresponding to a diameter of the outlet of the fan module **34** may be formed in the guide plate **51**, and a sleeve groove **511** in which the cover sleeve **54** may be seated may be formed on an upper surface of the guide plate **51** corresponding to an outside of the outlet **512**.

[0047] The pinion **53** may be located on an edge of the sleeve groove **511** and engaged with the rotation rack **541** formed on the outer peripheral surface of the cover sleeve **54**. As the rotation motor **52** operates, the pinion **53** engages with the rotation rack **541** and rotates, causing the cover sleeve **54** to rotate clockwise or counterclockwise. The cover sleeve **54** and the discharge cover **14** may rotate as one body. The discharge cover **14** may include a circular upper plate **141** and a lower plate **142** coupled to a bottom of the upper plate **141**. The lower plate **142** may be formed to be rounded to be convex downward from an edge to a center.

[0048] The tilting module may include an arc-shaped tilting neck **56** that is rounded at a predetermined curvature and has a tilting rack **561** formed on its upper surface, a tilting gear **57** engaged with the tilting rack **561**, a tilting motor **58** that rotates the tilting gear **57**, a bracket **59** that supports the tilting motor **58**, and a guide roller **60** that guides movement of the tilting neck **56** in a circumferential direction.

[0049] The bracket **59** may include a fixing bracket **591** and a support bracket **592**. The fixing bracket **591** is a bracket that connects a rear surface of the shield cover **55** and the rear surface of the cover sleeve **54**, and one or more fastening members may sequentially pass through fastening holes **542** formed on a back of the cover sleeve **54** and the fixing bracket **591** to be inserted into a seating protrusion **553** on the rear surface of the shielding cover **55**. Therefore, when the cover sleeve **54** rotates, the shielding cover **55** and the discharge cover **14** rotate together.

[0050] As shown in the drawings, the support bracket **592** may be orthogonally coupled to the fixing bracket **591**, and a shaft of the tilting motor **58** may pass through the support bracket **592**. The tilting gear **57** may be coupled to the shaft of the tilting motor **58** that has passed through the support bracket **592**.

[0051] When the tilting motor **58** is driven, the tilting gear **57** rotates to raise or lower the tilting neck **56**. The discharge cover **14** coupled to an upper end of the tilting neck **56** may be tilted at a predetermined angle according to a rotational amount of the tilting neck **56**. The upper end of the tilting neck **56** may be coupled to the lower plate **142** of the discharge cover **14** at a point spaced apart from a center of the lower plate **142** towards the rear surface of the shielding cover **55**.

[0052] FIG. **11** is a perspective view showing a combination of a shielding cover and the tilting module according to an embodiment. FIG. **12** is a longitudinal cross-sectional view, taken along line XII-XII of FIG. **11**, and FIG. **13** is a longitudinal cross-sectional view, taken along line XIII-XIII of FIG. **11**.

[0053] Referring to FIGS. **11** to **13**, the tilting neck **56** rotates in the vertical direction while being supported by the guide roller **60**. An outer surface (bottom surface) of the tilting neck **56** may be supported by outer guide roller **602**, and an inner surface (upper surface) may be supported by inner guide roller **601**. Additionally, the outer guide roller **602** may be provided as a pair at left and right or lateral edges of the tilting neck **56**. In addition, multiple pairs of outer guide roller **602** may be provided in a longitudinal direction of the tilting neck **56**. Like the outer guide roller **602**, multiple pairs of the inner guide rollers **601** may be provided on the inner surface of the tilting neck **56**.

[0054] The outer guide roller **602** may be connected to outer shaft **552** that protrudes from an inner surface of the shield cover **55** and contact the outer surface of the tilting neck **56**. Additionally, the inner guide roller **601** may be connected to inner shaft **551** provided in the shielding cover **55** and may contact the inner surface of the tilting neck **56**.

[0055] FIG. **14** is an enlarged view of portion A of FIG. **8**. Referring to FIG. **14**, a light-emitting module K and a sensor member S may be mounted on the discharge cover **14**, and at least a portion of the discharge cover **14** may be formed to be transparent to allow light irradiated from the light-emitting module K to be emitted to the outside. The sensor member S may include at least one of a temperature sensor that detects a temperature of air discharged through the main discharge port **131** and a humidity sensor that detects a humidity of air discharged through the main discharge port **131**. The light emitting module K may include an LED module that emits light of a specific color.

[0056] For example, only an edge portion of the discharge cover **14** may be formed to be transparent in a strip shape. A remaining portion may be molded to be opaque. The portion formed to be transparent may be defined as a transparent portion **143**. In addition, the light emitted from the light emitting module K may pass through the transparent portion **143** and be emitted to the outside of the discharge cover **14**, so that it may be viewed by a user's eyes.

[0057] In addition, a luminance of light emitted from the light emitting module K may be varied or a color of the light may be changed depending on the temperature or humidity of the discharged air detected by the sensor member S. For example, as the temperature of the discharge air increases, a red light may be emitted to the outside through the transparent portion **143**, and as the temperature decreases, a blue light may be emitted to the outside through the transparent portion **143**.

[0058] It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0059] It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

[0060] Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0061] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0062] Embodiments are described herein with reference to cross-section illustrations that are

schematic illustrations of idealized embodiments (and intermediate structures). As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

[0063] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0064] Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

[0065] Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A dehumidifier, comprising: a housing including a suction grill formed on a rear surface thereof, and a discharge port formed on an upper surface thereof; a dehumidification module provided inside of the housing, wherein the dehumidification module includes at least a compressor, a condenser, an evaporator, and a fan module; a water tank detachably provided on one side of the housing and that stores condensate generated during a dehumidification process; a discharge cover that opens and closes the discharge port; a tilting module connected to a bottom of the discharge cover inside of the housing to tilt the discharge cover; and a shielding cover that accommodates the tilting module, wherein the tilting module includes: a tilting motor fixed to the shielding cover; a tilting gear connected to a shaft of the tilting motor; and a tilting neck curved at a predetermined curvature and having on one surface thereof a tilting rack that engages with the tilting gear.
2. The dehumidifier according to claim 1, wherein the tilting module further comprises one or more inner guide rollers in contact with an upper surface of the tilting neck.
3. The dehumidifier according to claim 2, wherein the tilting module further comprises one or more outer guide rollers in contact with a lower surface of the tilting neck.
4. The dehumidifier according to claim 3, wherein the tilting module further comprises at least one bracket that couples the tilting motor to a rear surface of the shielding cover.
5. The dehumidifier according to claim 4, wherein the at least one bracket includes: a fixing bracket fixed to the rear surface of the shielding cover; and a support bracket coupled to a front surface of the fixing bracket, wherein a shaft of the tilting motor passes through the support bracket to be coupled to the tilting gear.

- 6.** The dehumidifier according to claim 5, further comprising a cover sleeve which is cylindrical and connects an outlet of the fan module and the discharge port, wherein the fixing bracket is fixed to the cover sleeve.
- 7.** The dehumidifier according to claim 6, wherein the discharge cover selectively shields an upper opening of the cover sleeve.
- 8.** The dehumidifier according to claim 1, wherein the discharge cover includes: a circular upper plate; a lower plate coupled to a bottom of the upper plate; and a light emitting module provided between the upper plate and the lower plate, wherein a transparent portion is configured to surround an edge of the discharge cover in a ring shape, such that light emitted from the light emitting module is discharged to an outside through the transparent portion.
- 9.** The dehumidifier according to claim 8, further comprising at least one of: a temperature sensor exposed to the lower plate to detect a temperature of air discharged through the discharge port; and a humidity sensor exposed to the lower plate to detect a humidity of air discharged through the discharge port.
- 10.** The dehumidifier according to claim 9, wherein a luminance or a color of the light emitted from the light emitting module is changed depending on the temperature of air detected by the temperature sensor or the humidity of air sensed by the humidity sensor.
- 11.** A dehumidifier, comprising: a housing including a suction inlet formed on a first surface thereof, and a discharge outlet formed on a second surface thereof; a dehumidification module provided inside of the housing, wherein the dehumidification module includes at least a compressor, a condenser, an evaporator, and a fan module; a discharge cover that opens and closes the discharge port; a tilting module connected to a bottom of the discharge cover inside of the housing to tilt the discharge cover; and a shielding cover that accommodates the tilting module, wherein the tilting module includes: a tilting motor fixed to the shielding cover; a tilting gear connected to a shaft of the tilting motor; and a tilting neck curved at a predetermined curvature and having on one surface thereof a tilting rack that engages with the tilting gear.
- 12.** The dehumidifier according to claim 11, wherein the tilting module further comprises one or more inner guide rollers in contact with an upper surface of the tilting neck.
- 13.** The dehumidifier according to claim 12, wherein the tilting module further comprises one or more outer guide rollers in contact with a lower surface of the tilting neck.
- 14.** The dehumidifier according to claim 13, wherein the tilting module further comprises at least one bracket that couples the tilting motor to a rear surface of the shielding cover.
- 15.** The dehumidifier according to claim 14, wherein the at least one bracket includes: a fixing bracket fixed to the rear surface of the shielding cover; and a support bracket coupled to a front surface of the fixing bracket, wherein a shaft of the tilting motor passes through the support bracket to be coupled to the tilting gear.
- 16.** The dehumidifier according to claim 15, further comprising a cover sleeve which is cylindrical and connects an outlet of the fan module and the discharge outlet, wherein the fixing bracket is fixed to the cover sleeve.
- 17.** The dehumidifier according to claim 16, wherein the discharge cover selectively shields an upper opening of the cover sleeve.
- 18.** The dehumidifier according to claim 1, wherein the discharge cover includes: a circular upper plate; a lower plate coupled to a bottom of the upper plate; and a light emitting module provided between the upper plate and the lower plate, wherein a transparent portion is configured to surround an edge of the discharge cover in a ring shape, such that light emitted from the light emitting module is discharged to an outside through the transparent portion.
- 19.** The dehumidifier according to claim 18, further comprising at least one of: a temperature sensor exposed to the lower plate to detect a temperature of air discharged through the discharge outlet; and a humidity sensor exposed to the lower plate to detect a humidity of air discharged through the discharge outlet.

20. The dehumidifier according to claim 19, wherein a luminance or a color of the light emitted from the light emitting module is changed depending on the temperature of air detected by the temperature sensor or the humidity of air sensed by the humidity sensor.
